

Aurora Felisari

M.Sc. STUDENT IN COMPUTER SCIENCE

Parma, Italy

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Education

Master's Student in Computer Science – AI Track (expected graduation Q3 2026)

University of Parma

2024 – PRESENT

Parma, Italy

I'm currently studying for my Master's Degree in Computer Science at the University of Parma, where I'm deepening my knowledge of advanced technologies and sharpening the skills I need to work on complex, real-world problems. The program is strongly oriented towards Artificial Intelligence, with a focus on its applications in industrial settings, Machine Learning, and Software Security. Alongside the core curriculum, I am also attending additional courses in *Autonomous Driving and ADAS Technologies* and *Virtual and Augmented Reality Systems*, which further broaden my expertise in intelligent systems and immersive computing.

Bachelor's Degree in Computer Science

University of Parma

2021 – 2024

Parma, Italy

I earned my Bachelor's Degree in Computer Science at the University of Parma, where I built a strong foundation in both the theory and practice of modern computing. The program gave me hands-on experience with current technologies, especially in the ICT field, and helped me understand how these tools are applied in real-world scenarios. It also taught me how to keep up with the fast pace of technological change — a skill I consider essential in this field.

Italian High School Diploma (Linguistic Track)

Liceo Linguistico Daniele Manin

2015 – 2020

Cremona, Italy

During my five years of high school, I deepened my understanding of grammar and studied the history and literature of the three languages I had chosen: English, German, and Spanish. The curriculum also included weekly conversation classes with native speakers.

Experience

Curricular Internship – Multi-Agent Conversational System

OT Consulting

MAR 2026 – JUL 2026

Reggio Emilia, Italy

Designed and developed an enterprise conversational bot built from scratch to support company employees in retrieving internal information and automating business workflows through natural language interaction. The system was architected as a multi-agent pipeline, with specialized agents capable of handling heterogeneous tasks with contextual awareness and dynamic decision-making. Took end-to-end ownership of the project, spanning system design, agent architecture, cloud deployment, and iterative refinement based on real use-case feedback. The project serves as the foundation for the Master's thesis.

Tutor – Summer Camp "Ragazze Digitali ER"

University of Parma

JUN 2025 – JUL 2025

Parma, Italy

Guided and supported a group of 16-to-18-year-old girls with no prior programming experience during an intensive course introducing the fundamentals of computer science. The program aimed to empower participants through accessible and inclusive digital education. Taught the basics of Python programming and assisted in the development of original video games combining code and graphic design, enabling students to apply newly acquired concepts in practical, creative ways. Acted as a constant point of reference throughout the course, offering technical guidance, motivation, and mentoring in both group and individual settings.

Undergraduate Internship – Cognitive Rehabilitation with SAR

University of Parma

NOV 2023 – SEP 2024

Parma, Italy

Carried out an interdisciplinary internship in collaboration with the Department of Psychobiology and Cognitive Neuroscience. Worked on cognitive rehabilitation for patients with Mild Cognitive Impairment (MCI), focusing on short-term memory disorders. Contributed to the development of software for therapeutic sessions supported by the NAO Socially Assistive Robot (SAR), integrating adaptive interaction and task personalization. The internship laid the groundwork for my Bachelor's thesis project.

Academic Activity

Autonomous Driving and ADAS Technologies

SEP 2025–

University of Parma

Parma, Italy

Perception, Vision and Autonomous Driving Systems: Attending a course dedicated to the study of technologies for self-driving vehicles and Advanced Driver Assistance Systems (ADAS). The program covers perception and artificial vision, sensor technologies, mapping and simulation environments, planning and control algorithms, and system architectures. The course integrates theoretical lectures with mandatory laboratory activities in perception, requiring development skills in C/C++. External seminars complement the training with case studies from research and industry.

Virtual and Augmented Reality Systems

SEP 2025 – JUL 2026

University of Parma

Parma, Italy

Design and Development of VR/AR Applications: Attended a course dedicated to modern Virtual and Augmented Reality technologies for interactive 3D simulated environments. The program covers interaction paradigms, input devices (haptic, depth cameras, motion tracking), stereoscopic displays, and collision detection algorithms. Includes extensive laboratory work on Unity3D with C# programming, physics-based simulation, and Bullet Physics in C++. As the final individual project, designed and developed from scratch a low-poly VR cooking simulation game for Meta Quest 3, inspired by *Cooking Mama*, applying the theoretical knowledge acquired throughout the course. The game features immersive hand-tracking interactions, physics-based ingredient manipulation, and 3D assets modeled in Blender. Built using Unity3D with C# and integrated with the Meta XR SDK.

Introduction to Machine Learning

MAR 2026 – JUN 2026

University of Parma

Parma, Italy

Le statistiche bastano? – Statistical Analysis of Pokémon Data: Attended a course on statistical learning methods, covering probability theory, dimensionality reduction, regression and classification, regularization, model selection, and hypothesis testing. The final examination project is a statistical analysis of a public dataset of around 800 Pokémon, investigating whether elemental type, legendary status, and generation can be predicted from the six base stats. Produced 21 visualizations (heatmaps, KDE plots, Q-Q plots, boxplots) using Matplotlib and Seaborn.

Development of Reliable, Safe and Secure Software Project

JAN 2025– JUL 2025

University of Parma

Parma, Italy

Isolette Thermostat Control System: Designed and implemented a safety-critical thermostat control system for a neonatal incubator (Isolette), within a university project developed in close collaboration with the software verification company BUGSENG. The project served as a concrete case study to explore the full software development lifecycle under industry-grade safety and reliability constraints. Starting from requirements specification and system design, we carried out all development stages: coding in C, producing documentation, conducting verification activities, and constructing structured safety arguments, all in line with the guidelines of functional safety standards for critical embedded systems. The course emphasized real-world industrial practices, focusing on the cost-benefit trade-offs of formal, semi-formal, and informal methods. It also highlighted the importance of traceability, role separation (developers, verifiers, assessors), and rigorous communication. We used Git for version control and Docker to manage a shared development environment throughout the entire process.

Fundamentals of Artificial Intelligence

FEB 2025– JUN 2025

University of Parma

Parma, Italy

From zero to continuity: Benchmarking ReLU, Leaky ReLU & GELU: Studied and compared ReLU, Leaky ReLU and GELU activation functions for feedforward neural networks. Implemented both single-hidden-layer and multi-hidden-layer MLP architectures using PyTorch. Benchmarked performance on MNIST and FashionMNIST classification tasks, analyzing accuracy, convergence speed and generalization. Documented the impact of each activation on training dynamics and final model performance.

Quantum Computing Project

JAN 2025– FEB 2025

University of Parma

Parma, Italy

Solving Sudoku with Grover's Algorithm: Explored quantum search algorithms using Qiskit. Implemented Grover's Algorithm to identify solutions in an unstructured search space, analyzing its quadratic speed-up compared to classical approaches. Collaborated in the presentation and critical analysis of circuit behavior and performance.

Declarative Programming Project

JAN 2025– FEB 2025

University of Parma

Parma, Italy

Answer Set Programming for Gerrymandering: Model, Benchmarks and Evaluation: Designed and implemented an Answer Set Programming model to solve the Gerrymandering problem over voting grids. Generated 100 benchmark instances with varying parameters and evaluated solver performance under time constraints. Produced a structured report with model design, optimization strategies, and experimental results.

This thesis details the development of a cognitive rehabilitation system using Socially Assistive Robotics (SAR) and Explainable AI (XAI). The system employs a NAO humanoid robot for personalized therapeutic sessions for patients with mild memory disorders. It features adaptive cognitive tasks modeled with Answer Set Programming (ASP) and adheres to EU AI Act guidelines for transparency. The project involved collaboration with psychologists and aims to make therapy more engaging and adaptable.

Languages

- Native speaker of **Italian**.
- Strong command of **English** (B2), confidently used in academic and professional settings.
- Intermediate knowledge of **German** and **Spanish** (B1), acquired through high school education and international experiences.

Skills and Additional Information

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C, C++, C#, MATLAB, ASP (Answer Set Programming)
- **Libraries & Frameworks:** PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, Matplotlib, Seaborn, LangChain, Qiskit, Clingo
- **Technologies & Tools:** Git, Jupyter Notebook, Docker, PostgreSQL, MySQL, Figma, Unity3D, Vertex AI, GCP
- **Systems & Architecture:** Operating Systems, Network Protocols, Parallel and High-Performance Computing
- **Data Management:** Relational Databases, Information Systems, Database Design and Querying (SQL)
- **Software Engineering:** Agile Development, Version Control with GitHub, Model-Based Design (Isolette Thermostat Project)
- **Office Suite:** Microsoft Word, Excel, PowerPoint (Advanced proficiency)

DOMAINS & APPLICATIONS

- **Virtual & Augmented Reality:** Unity3D development, C# programming, Physics-based simulation, Meta XR SDK, Blender
- **Autonomous Driving & ADAS:** Sensor technologies, Computer Vision, Perception algorithms, Planning & Control
- **Statistical Analysis & Machine Learning:** Probabilistic modeling, Dimensionality reduction (PCA, Factor Analysis), Regression and classification, Regularization (Ridge, Lasso), Hypothesis testing (t-test, ANOVA, χ^2 , Fisher)

SOFT SKILLS

- **Autonomy & Ownership:** Repeatedly took end-to-end responsibility for projects from design to delivery, working independently with minimal supervision
- **Interdisciplinary Collaboration:** Worked across domain boundaries with psychologists and neuroscientists (thesis) and with a software verification company (BUGSENG)
- **Technical Communication:** Produced structured reports, formal documentation, and project presentations for both academic and industry audiences
- **Problem-Solving Under Constraints:** Worked within real-world safety standards (Isolette/BUGSENG), benchmark time limits (ASP), and iterative industry feedback cycles (OT Consulting)

OTHER INFORMATION

- **Driving License:** Category B – Full, valid in the EU